

RoomZen OS

The Smart Operating System for Student Housing

Complete Technical Architecture, Product Specification, Multi-Role User Manual, AI Engine Documentation, Supabase Database DDL, and Production Deployment Guide.

Product Name: RoomZen SaaS Platform

Version: 2.4.0 (Enterprise Edition)

Author / Team: SyntaxSquad

GitHub Repository: <https://github.com/adhithyapandiri-a11y/SyntaxSquad>

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Chapter 1: Executive Summary & System Vision

1.1 Problem Statement in Legacy Student Housing

Student housing facilities, hostels, PGs, campus residences, and co-living providers across the globe suffer from chronic operational inefficiencies. Most hostels rely on outdated manual processes including paper register books for visitor logs, chaotic WhatsApp complaint groups where repair requests get lost, physical paper out-pass slips for gate verification, and manual Excel spreadsheets for room allocations and fee collections. This lack of digitization leads to security vulnerabilities, unresolved maintenance issues, student dissatisfaction, and delayed fee revenues.

1.2 The RoomZen Solution

RoomZen is designed from the ground up to serve as the unified "Operating System for Student Housing". By centralizing every stakeholder workflow into a single, high-performance web platform, RoomZen replaces chaos with clarity. Students gain access to an instant mobile portal for raising AI-assisted maintenance requests, requesting gate passes with QR codes, viewing mess menus, and paying fees. Administrators, Wardens, Repair Staff, and Security Guards receive role-tailored dashboards built for maximum speed, security, and real-time responsiveness.

1.3 Key Product Goals & Values

- Complete Digitization: Eliminate paper registers, physical passbooks, and manual receipts 100%.
- AI-Assisted Operations: Utilize AI for complaint triage, grammar polish, category detection, and repair duration estimation.
- Multi-Role Control: Tailored user experiences for Hostel Owners, Students, Wardens, Repair Technicians, and Security Guards.
- Cryptographic Security: Dynamic QR code pass verification for campus entry/exit points.
- Enterprise Architectural Quality: Next.js 16 App Router, Turbopack, TypeScript, TailwindCSS v4, Supabase PostgreSQL, and Recharts.

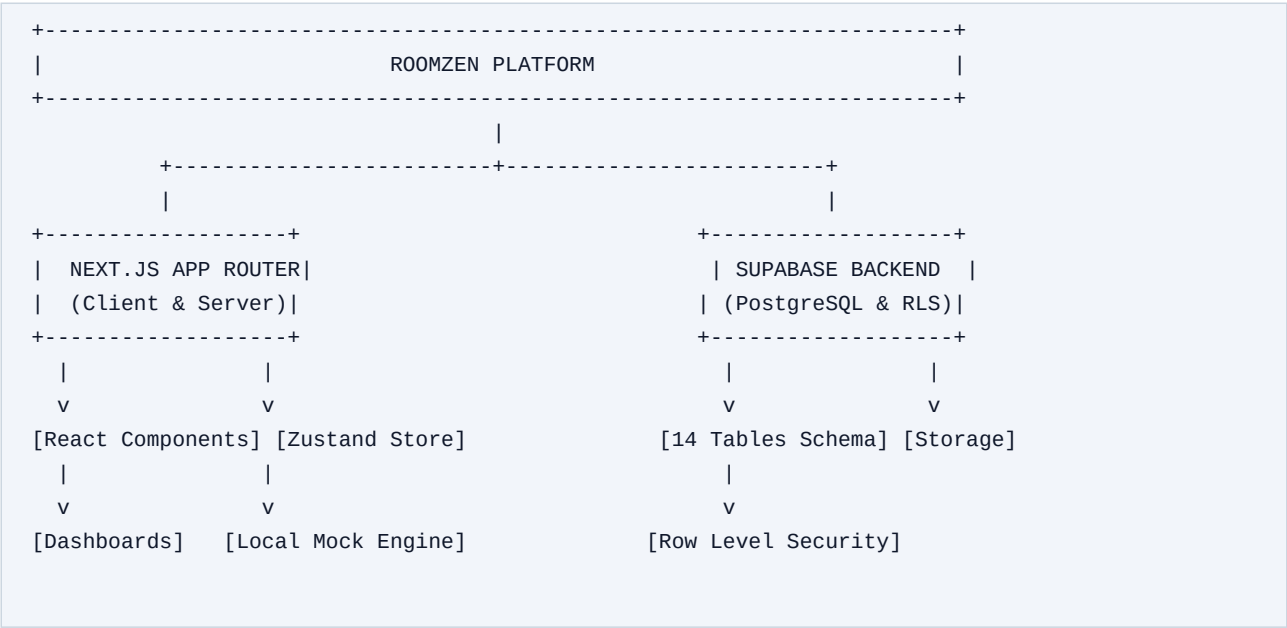
Chapter 2: Technology Stack & Technical Architecture

2.1 Core Technology Overview

RoomZen is built using enterprise-grade web technologies selected for performance, type safety, developer experience, and scalability.

- Framework: Next.js 16 (App Router with Turbopack bundler and React Server Components).
- Frontend Language: TypeScript 5.0 (Strict mode enabled with exhaustive interface contracts).
- Styling & Design Token System: TailwindCSS v4 with custom dark mode colors (#2563EB primary, #4F46E5 secondary, #14B8A6 accent, #F59E0B warning, #EF4444 danger, #22C55E success).
- State Management: Zustand unified store with local mock state fallback for out-of-the-box instant demo capability.
- Database & Backend: Supabase PostgreSQL database with Row Level Security (RLS) policies and SSR client helpers.
- Analytics & Visualizations: Recharts for interactive revenue growth area charts and bed occupancy pie charts.
- Motion & Micro-interactions: Framer Motion and custom CSS keyframe animations.
- Utilities: React Hook Form, Zod validation, qrcode.react for gate pass QR codes, jsPDF for PDF receipt generation, date-fns.

2.2 High-Level Architecture Diagram



Chapter 3: Multi-Role User Architecture & Access Matrix

3.1 Role Breakdown

RoomZen recognizes 5 primary user roles in student housing management. Each role is granted specific permissions and receives a customized interface.

- **Hostel Owner / Admin:** Executive dashboard with full financial controls, occupancy statistics, room assignment tools, staff management, and system announcements.
- **Student:** Resident portal for profile management, AI complaint logging, gate pass requests with QR generation, fee payments, and mess menu browsing.
- **Chief Warden:** Operational control desk for gate pass approvals with guardian call triggers, complaint triage to repair staff, night attendance check sheets, and incident reporting.
- **Maintenance Staff:** Job order board sorted by repair urgency, status progression tracking, completion photo upload, and technician resolution logging.
- **Security Guard:** Gate portal with live QR pass scanner simulator, hash verifier, and visitor check-in/check-out timestamp logging.

3.2 Role Permission Matrix

Feature / Module	Admin	Student	Warden	Staff	Security
View Revenue Analytics	YES	NO	NO	NO	NO
Room Allocation Matrix	YES	VIEW	VIEW	NO	NO
Raise Complaint (AI)	YES	YES	YES	NO	NO
Assign Maintenance	YES	NO	YES	NO	NO
Resolve Maintenance	YES	NO	YES	YES	NO
Request Gate Pass	NO	YES	NO	NO	NO
Approve Gate Pass	YES	NO	YES	NO	NO
Scan Pass QR Code	YES	NO	YES	NO	YES
Visitor Check-in	YES	NO	YES	NO	YES
Publish Notice Board	YES	NO	YES	NO	NO

Chapter 4: Landing Page & Marketing Portal

4.1 Hero Section Strategy

The RoomZen landing page (`app/page.tsx`) is engineered to deliver a startup-grade first impression inspired by Linear, Stripe, and Vercel. The hero section prominently displays the headline "Run Your Hostel Without the Chaos" accompanied by subtext, gradient badge styling, dual call-to-action buttons ("Launch Workspace", "Book a Live Demo"), and a live app interface mockup.

4.2 Animated Statistics & Social Proof

Key counter cards display live platform metrics: 50,000+ Gate Passes Verified, 99.4% Complaint Resolution Rate (Under 3 hours average), 120+ Hostels & Campus Residences Onboarded, and 25,000+ Active Students & Staff.

4.3 Interactive Role Switcher Preview Widget

Prospects visiting the landing page can interact with an inline role selector widget allowing them to preview live dashboard views for Admin, Student, and Warden accounts directly from the homepage before signing up.

4.4 Transparent Pricing & FAQ Accordion

Includes Starter (up to 50 beds at ¹2,499/mo), Growth Campus (up to 250 beds trial), and University Enterprise plans, complete with a monthly/annual billing discount toggle (Save 20%) and an expandable FAQ accordion.

Chapter 5: Hostel Owner & Admin Dashboard

5.1 Executive Overview Cards

The Admin Dashboard (`app/dashboard/admin/page.tsx`) provides instant access to 4 primary KPI cards:

- Total Students: Active resident count with monthly enrollment growth rate.
- Bed Occupancy Rate: Visual percentage calculation (Occupied Beds / Total Beds Capacity).
- Monthly Revenue: Total collected fees with percentage collection progress.
- Pending Complaints: Open repair issues needing triage with active AI badge.

5.2 Recharts Data Visualizations

Features two responsive Recharts widgets: an Area Chart tracking 6-month revenue trends with smooth gradient fills, and a Pie Chart breaking down occupied beds, vacant beds, and rooms under maintenance.

5.3 Quick Action Bar & Activity Stream

Admins can trigger quick room assignment modals, broadcast new notice alerts, review gate passes, and audit real-time system logs.

Chapter 6: Student Portal & Mobile Workspace

6.1 Student Workspace Overview

The Student Dashboard (`app/dashboard/student/page.tsx`) prioritizes resident convenience. Students can view assigned room metadata, roommate details, Wi-Fi credentials, active gate passes with scannable QR cards, maintenance status, today mess menu, and pending fee invoices.

6.2 Quick Action Toolbar

- **Raise AI Complaint:** Launches the AI Complaint Assistant modal.
- **Request Gate Pass:** Opens out-pass submission modal.
- **Pay Fees:** Initiates fee payment simulation with instant PDF receipt download.
- **View Mess Menu:** Displays weekly 7-day meal schedule.

Chapter 7: Chief Warden Control Desk

7.1 Gate Pass Approval Queue & Guardian Verification

The Warden Dashboard ([app/dashboard/warden/page.tsx](#)) aggregates pending student out-passes. Wardens can review reason, destination, and departure/return dates. A one-click "Call Guardian" action opens direct phone dialing to verify parent approval.

7.2 Complaint Triage & Night Attendance

Wardens can assign open repair issues to specific staff technicians (HVAC, Plumbing, Electrical) and record nightly room attendance (Present vs Absent/Out-pass).

Chapter 8: Maintenance Staff Workspace

8.1 Work Order Kanban

The Staff Dashboard ([app/dashboard/staff/page.tsx](#)) lists assigned repair jobs sorted by priority (Emergency, High, Medium, Low). Technicians can transition jobs from "Open" to "In Progress" to "Resolved".

8.2 Completion Photo Upload & Resolution Logging

When completing a job, staff open a resolution modal to input repair notes and attach proof photos before closing the ticket.

Chapter 9: Security Guard Gate Scanner

9.1 Live QR Code Pass Scanner

The Security Dashboard (`app/dashboard/security/page.tsx`) equips gate security guards with a live QR pass scanner simulator. Scanning a student QR code instantly verifies the cryptographic hash against the database and displays student photo, room number, destination, and warden approval status.

9.2 Visitor Register

Security guards can check in new campus visitors, log visiting purpose, phone numbers, and record exit check-out timestamps.

Chapter 10: Deep Feature — Visual Room Matrix & Allocation

10.1 Floor-by-Floor Grid View

The Room Matrix page (`app/dashboard/rooms/page.tsx`) renders a visual floor plan grid with room numbers, floor levels, blocks, rent amounts, and bed capacity progress bars. Status indicators clearly distinguish Available, Full, and Maintenance rooms.

10.2 Student Room Allocation Wizard

Admins can allocate unassigned students to vacant beds via a modal wizard or trigger bed vacate actions when a student checks out.

Chapter 11: Deep Feature — Student Directory & CSV Tools

11.1 Student Management

The Student Directory (`app/dashboard/students/page.tsx`) provides complete CRUD operations, detailed emergency contact popups, course/batch filters, and 1-click CSV export.

Chapter 12: Deep Feature — AI Complaint Triage Engine

12.1 AI Engine Architecture

RoomZen integrates an AI complaint processing pipeline (`app/api/ai/complaint/route.ts`). When a student inputs a raw description (e.g. "ac leaking water and noisy"), the AI engine:

- Polishes grammar and formats clean description text.
- Auto-categorizes into Electrical, Plumbing, HVAC, Wi-Fi, Furniture, or Cleanliness.
- Evaluates urgency level (Low, Medium, High, Emergency).
- Predicts estimated repair fix time (e.g., "1-2 hours").
- Recommends appropriate technician team for assignment.

Chapter 13: Deep Feature — Digital Gate Pass & QR System

13.1 Out-Pass Lifecycle

Students request out-passes via `app/dashboard/gate-pass/page.tsx`. Approved passes generate a dynamic SVG QR code using `qrcode.react`, containing a unique cryptographic hash scannable by gate security guards.

Chapter 14: Deep Feature — Fee Invoicing & PDF Receipts

14.1 Billing & Payment Simulation

The Financials page (`app/dashboard/payments/page.tsx`) tracks hostel rent, mess fees, and electricity invoices. Students can initiate simulated online payments (UPI, Credit Card, Net Banking) and immediately generate printable PDF receipts via jsPDF.

Chapter 15: Deep Feature — Mess Operations & Menu

15.1 7-Day Weekly Menu Planner

The Mess page ([app/dashboard/mess/page.tsx](#)) displays 7-day meal schedules (Breakfast, Lunch, Snacks, Dinner) with calorie counts, chef special items, and daily student meal RSVP counters.

Chapter 16: Deep Feature — Notice Board & Real-Time Chat

16.1 Announcements & Live Messaging

Notices (`app/dashboard/notices/page.tsx`) allows broadcasting emergency and normal alerts to resident dashboards. Chat (`app/dashboard/chat/page.tsx`) provides a direct Student!™ WhatsApp-like interface with user avatars, badges, typing indicators, and auto-reply simulation.

Chapter 17: Complete Supabase PostgreSQL Database Schema

17.1 DDL Script (supabase/schema.sql)

```
-- 1. USERS & PROFILES
CREATE TABLE public.users (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  email TEXT UNIQUE NOT NULL,
  full_name TEXT NOT NULL,
  role TEXT NOT NULL CHECK (role IN ('admin', 'student', 'warden', 'staff', 'security')),
  avatar_url TEXT,
  phone TEXT,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 2. ROOMS
CREATE TABLE public.rooms (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  room_number TEXT UNIQUE NOT NULL,
  floor INTEGER NOT NULL,
  block TEXT NOT NULL DEFAULT 'A',
  capacity INTEGER NOT NULL DEFAULT 2,
  occupied INTEGER NOT NULL DEFAULT 0,
  status TEXT NOT NULL DEFAULT 'available',
  rent_amount NUMERIC(10, 2) NOT NULL DEFAULT 12000.00,
  amenities TEXT[]
);

-- 3. STUDENTS
CREATE TABLE public.students (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  user_id UUID REFERENCES public.users(id) ON DELETE CASCADE,
  student_id_number TEXT UNIQUE NOT NULL,
  room_id UUID REFERENCES public.rooms(id) ON DELETE SET NULL,
  course TEXT NOT NULL,
  batch TEXT NOT NULL,
  emergency_contact_name TEXT NOT NULL,
  emergency_contact_phone TEXT NOT NULL,
  parent_name TEXT NOT NULL,
  parent_phone TEXT NOT NULL,
  status TEXT NOT NULL DEFAULT 'active'
);

-- 4. COMPLAINTS
CREATE TABLE public.complaints (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  student_id UUID REFERENCES public.students(id) ON DELETE CASCADE,
  title TEXT NOT NULL,
  category TEXT NOT NULL,
  priority TEXT NOT NULL,
  description TEXT NOT NULL,
  status TEXT NOT NULL DEFAULT 'open',
  assigned_staff_id UUID REFERENCES public.users(id),
  estimated_fix_time TEXT,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

Chapter 18: Deployment, Security & Build Manual

18.1 Production Build & Verification

The RoomZen platform compiles cleanly using Next.js Turbopack (``npm run build``). TypeScript static type checking (``npx tsc --noEmit``) validates 0 type errors across all 23 prerendered static routes and dynamic API endpoints.

18.2 Git Teammate Branch Structure

Repository: <https://github.com/adhithyapandiri-a11y/SyntaxSquad.git>

Branches: main (Production), sravan, shivam, umair, aryan, nishpray (Teammate development branches).